

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                      |                                                |
|----------------------|------------------------------------------------|
| Product Description: | <u>Hydrazine hydrate, 80% (Hydrazine, 51%)</u> |
| Product Grade:       | SQ                                             |
| Cat No. :            | Q2494G                                         |
| CAS No               | 10217-52-4                                     |
| Molecular Formula    | H4 N2 . x H2 O                                 |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

### 1.3. Details of the supplier of the safety data sheet

|         |                                                                                                                                  |
|---------|----------------------------------------------------------------------------------------------------------------------------------|
| Company | Thermo Fisher Scientific India Pvt. Ltd<br>403-404, B-wing, Delphi,<br>Hiranandani Business Park,<br>Powai, Mumbai 400076, INDIA |
|---------|----------------------------------------------------------------------------------------------------------------------------------|

|                |                                     |
|----------------|-------------------------------------|
| E-mail address | laboratorysolution@thermofisher.com |
|----------------|-------------------------------------|

### 1.4. Emergency telephone number

India Toll Free: 18 00 22 22 30  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001 (202) 483-7616

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

##### Physical hazards

Based on available data, the classification criteria are not met

##### Health hazards

|                                    |                     |
|------------------------------------|---------------------|
| Acute oral toxicity                | Category 3 (H301)   |
| Acute dermal toxicity              | Category 3 (H311)   |
| Acute Inhalation Toxicity - Vapors | Category 2 (H330)   |
| Skin Corrosion/Irritation          | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation  | Category 1 (H318)   |

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|                                                    |                                         |
|----------------------------------------------------|-----------------------------------------|
| Skin Sensitization<br>Carcinogenicity              | Category 1 (H317)<br>Category 1B (H350) |
| <b>Environmental hazards</b>                       |                                         |
| Acute aquatic toxicity<br>Chronic aquatic toxicity | Category 1 (H400)<br>Category 1 (H410)  |

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### Hazard Statements

H330 - Fatal if inhaled  
H314 - Causes severe skin burns and eye damage  
H317 - May cause an allergic skin reaction  
H350 - May cause cancer  
H410 - Very toxic to aquatic life with long lasting effects  
H301 + H311 - Toxic if swallowed or in contact with skin  
Combustible liquid

### Precautionary Statements

P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P302 + P350 - IF ON SKIN: Gently wash with plenty of soap and water  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
P310 - Immediately call a POISON CENTER or doctor/physician

### Additional EU labelling

Restricted to professional users

## 2.3. Other hazards

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component | CAS No | EC No | Weight % | CLP Classification - According to<br>GB-CLP Regulations UK SI 2019/720 and<br>UK SI 2020/1567 |
|-----------|--------|-------|----------|-----------------------------------------------------------------------------------------------|
|-----------|--------|-------|----------|-----------------------------------------------------------------------------------------------|

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|                     |            |                   |    |                                                                                                                                                                                                                                       |
|---------------------|------------|-------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydrazine (hydrate) | 10217-52-4 |                   | 80 | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 2 (H330)<br>Skin Corr. 1B (H314)<br>Skin Sens. 1 (H317)<br>Eye Dam. 1 (H318)<br>Carc. 1B (H350)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410)                        |
| Hydrazine           | 302-01-2   | EEC No. 206-114-9 | -  | Flam. Liq. 3 (H226)<br>Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 2 (H330)<br>Skin Corr. 1B (H314)<br>Skin Sens. 1 (H317)<br>Eye Dam. 1 (H318)<br>Carc. 1B (H350)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) |
| Water               | 7732-18-5  | 231-791-2         | 20 | -                                                                                                                                                                                                                                     |

| Component | Specific concentration limits (SCL's) | M-Factor | Component notes |
|-----------|---------------------------------------|----------|-----------------|
| Hydrazine | -                                     | 10       | -               |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Difficulty in breathing. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or

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emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

## 4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Combustible material. Containers may explode when heated. Do not allow run-off from fire-fighting to enter drains or water courses.

#### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Ammonia, Hydrogen.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### 6.3. Methods and material for containment and cleaning up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

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Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition.

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 7.2. Conditions for safe storage, including any incompatibilities

Keep away from heat, sparks and flame. Store under an inert atmosphere. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

Technical Rules for Hazardous Substances (TRGS) 510 Class 6.1A  
Storage Class (LGK) (Germany)

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE** - 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

| Component | The United Kingdom                                                                                                                        | European Union                                                  | Ireland                                                                                                                            |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Hydrazine | STEL: 0.03 ppm 15 min<br>STEL: 0.039 mg/m <sup>3</sup> 15 min<br>TWA: 0.01 ppm 8 hr<br>TWA: 0.013 mg/m <sup>3</sup> 8 hr<br>Carc.<br>Skin | TWA: 0.013 mg/m <sup>3</sup> (8h)<br>TWA: 0.01 ppm (8h)<br>Skin | TWA: 0.01 ppm 8 hr.<br>TWA: 0.013 mg/m <sup>3</sup> 8 hr.<br>STEL: 0.03 ppm 15 min<br>STEL: 0.039 mg/m <sup>3</sup> 15 min<br>Skin |

#### Biological limit values

List source(s):

#### Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

No information available

#### Predicted No Effect Concentration (PNEC)

No information available.

### 8.2. Exposure controls

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## Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

## Personal protective equipment

### Eye Protection

Goggles (European standard - EN 166)

### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

### Skin and body protection

Long sleeved clothing.

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Particulates filter conforming to EN 143 or Inorganic gases and vapours filter Type B Grey conforming to EN14387

### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                          |                    |                       |
|--------------------------|--------------------|-----------------------|
| Physical State           | Liquid             |                       |
| Appearance               | Colorless          |                       |
| Odor                     | Ammonia-like       |                       |
| Odor Threshold           | No data available  |                       |
| Melting Point/Range      | -57 °C / -70.6 °F  |                       |
| Softening Point          | No data available  |                       |
| Boiling Point/Range      | 117.2 °C / 243 °F  |                       |
| Flammability (liquid)    | Combustible liquid | On basis of test data |
| Flammability (solid,gas) | Not applicable     | Liquid                |
| Explosion Limits         | Lower 4.8 Vol%     |                       |

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|                                         |                                    |                                   |
|-----------------------------------------|------------------------------------|-----------------------------------|
| Flash Point                             | Upper 100 Vol%<br>91 °C / 195.8 °F | Method - No information available |
| Autoignition Temperature                | 310 °C / 590 °F                    |                                   |
| Decomposition Temperature               | No data available                  |                                   |
| pH                                      | 12                                 | 640 g/l aq.sol                    |
| Viscosity                               | 1.33 mPa.s at 20 °C                |                                   |
| Water Solubility                        | Miscible                           |                                   |
| Solubility in other solvents            | No information available           |                                   |
| Partition Coefficient (n-octanol/water) |                                    |                                   |
| Component                               | log Pow                            |                                   |
| Hydrazine                               | -0.16                              |                                   |
| Vapor Pressure                          | 13 mbar @ 20 °C                    |                                   |
| Density / Specific Gravity              | 1.028                              |                                   |
| Bulk Density                            | Not applicable                     | Liquid                            |
| Vapor Density                           | 1.1 @ 15 °C                        | (Air = 1.0)                       |
| Particle characteristics                | Not applicable (liquid)            |                                   |

## 9.2. Other information

|                      |                                        |
|----------------------|----------------------------------------|
| Molecular Formula    | H4 N2 . x H2 O                         |
| Molecular Weight     | 32.04                                  |
| Explosive Properties | explosive air/vapour mixtures possible |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Do not allow evaporation to dryness. Air sensitive.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Exposure to air. Incompatible products. Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Acids. Bases. Finely powdered metals. Halogens. nitrogen oxides (NOx). Organic materials. Peroxides. Lead. Metals. copper. Butyl rubber.

### 10.6. Hazardous decomposition products

Nitrogen oxides (NOx). Ammonia. Hydrogen.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

|                     |            |
|---------------------|------------|
| (a) acute toxicity; |            |
| Oral                | Category 3 |
| Dermal              | Category 3 |

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Inhalation

Category 2

## Toxicology data for the components

| Component | LD50 Oral               | LD50 Dermal                | LC50 Inhalation                              |
|-----------|-------------------------|----------------------------|----------------------------------------------|
| Hydrazine | LD50 = 60 mg/kg ( Rat ) | LD50 = 91 mg/kg ( Rabbit ) | 570 ppm ( Rat ) 4 h<br>0.75 mg/L ( Rat ) 4 h |
| Water     | -                       | -                          | -                                            |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory No data available  
Skin Category 1

No information available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; Category 1B

May cause cancer The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | EU           | UK | Germany | IARC     |
|-----------|--------------|----|---------|----------|
| Hydrazine | Carc Cat. 1B |    | Cat. 2  | Group 2A |

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

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## Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Fish                                                                                                                                                                                                                                                                                        | Water Flea | Freshwater Algae                                                                                                                                                                                 |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydrazine | LC50: 0.28 - 1.34 mg/L, 96h static (Poecilia reticulata)<br>LC50: 1.81 - 2.79 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 1.17 mg/L, 96h (Lepomis macrochirus)<br>LC50: 0.54 - 1.31 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 0.7 - 1.3 mg/L, 96h flow-through (Lepomis macrochirus) |            | EC50: = 0.006 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: = 0.071 mg/L, 72h (Pseudokirchneriella subcapitata)<br>EC50: = 0.02 mg/L, 96h static (Pseudokirchneriella subcapitata) |

| Component           | Microtox                                                                     | M-Factor |
|---------------------|------------------------------------------------------------------------------|----------|
| Hydrazine (hydrate) | EC50 = 0.01 mg/L 15 min<br>EC50 = 0.01 mg/L 20 min<br>EC50 = 0.02 mg/L 5 min |          |
| Hydrazine           | EC50 = 0.01 mg/L 15 min<br>EC50 = 0.01 mg/L 20 min<br>EC50 = 0.02 mg/L 5 min | 10       |

## 12.2. Persistence and degradability

### Persistence

Miscible with water, Persistence is unlikely, based on information available.

### Degradability

Not relevant for inorganic substances.

### Degradation in sewage treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Hydrazine | -0.16   | No data available             |

## 12.4. Mobility in soil

The product is water soluble, and may spread in water systems . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

## 12.5. Results of PBT and vPvB assessment

No data available for assessment.

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

### Persistent Organic Pollutant

This product does not contain any known or suspected substance

### Ozone Depletion Potential

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

## 13.1. Waste treatment methods

### Waste from Residues/Unused

Should not be released into the environment. Waste is classified as hazardous. Dispose of

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|                                       |                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Products</b>                       | in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                                                                                                                        |
| <b>Contaminated Packaging</b>         | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                                        |
| <b>European Waste Catalogue (EWC)</b> | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                                         |
| <b>Other Information</b>              | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Solutions with high pH-value must be neutralized before discharge. Do not let this chemical enter the environment. |

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                                         |                             |
|-----------------------------------------|-----------------------------|
| <b>14.1. UN number</b>                  | UN2030                      |
| <b>14.2. UN proper shipping name</b>    | HYDRAZINE, AQUEOUS SOLUTION |
| <b>14.3. Transport hazard class(es)</b> | 8                           |
| <b>Subsidiary Hazard Class</b>          | 6.1                         |
| <b>14.4. Packing group</b>              | II                          |

### ADR

|                                         |                            |
|-----------------------------------------|----------------------------|
| <b>14.1. UN number</b>                  | UN2030                     |
| <b>14.2. UN proper shipping name</b>    | HYDRAZINE AQUEOUS SOLUTION |
| <b>14.3. Transport hazard class(es)</b> | 8                          |
| <b>Subsidiary Hazard Class</b>          | 6.1                        |
| <b>14.4. Packing group</b>              | II                         |

### IATA

|                                         |                             |
|-----------------------------------------|-----------------------------|
| <b>14.1. UN number</b>                  | UN2030                      |
| <b>14.2. UN proper shipping name</b>    | HYDRAZINE, AQUEOUS SOLUTION |
| <b>14.3. Transport hazard class(es)</b> | 8                           |
| <b>Subsidiary Hazard Class</b>          | 6.1                         |
| <b>14.4. Packing group</b>              | II                          |

|                                    |                                                                                                          |
|------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>14.5. Environmental hazards</b> | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
|------------------------------------|----------------------------------------------------------------------------------------------------------|

|                                           |                                  |
|-------------------------------------------|----------------------------------|
| <b>14.6. Special precautions for user</b> | No special precautions required. |
|-------------------------------------------|----------------------------------|

|                                                                      |                                |
|----------------------------------------------------------------------|--------------------------------|
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods |
|----------------------------------------------------------------------|--------------------------------|

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

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| Component           | CAS No     | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|---------------------|------------|-----------|--------|-----|-------|------|----------|------|------|
| Hydrazine (hydrate) | 10217-52-4 | -         | -      | -   | X     | X    | -        | X    | X    |
| Hydrazine           | 302-01-2   | 206-114-9 | -      | -   | X     | X    | KE-19981 | X    | X    |
| Water               | 7732-18-5  | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |

| Component           | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|---------------------|------------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Hydrazine (hydrate) | 10217-52-4 | -    | -                                             | -   | -    | -    | X     | -     |
| Hydrazine           | 302-01-2   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Water               | 7732-18-5  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component           | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------|------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Hydrazine (hydrate) | 10217-52-4 | -                                                                   | -                                                                                                                                        | -                                                                                                     |
| Hydrazine           | 302-01-2   | -                                                                   | Use restricted. See item 28.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | SVHC Candidate list - 206-114-9 - Carcinogenic, Article 57a                                           |
| Water               | 7732-18-5  | -                                                                   | -                                                                                                                                        | -                                                                                                     |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

### REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Seveso III Directive (2012/18/EC)

| Component           | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Hydrazine (hydrate) | 10217-52-4 | Not applicable                                                                            | Not applicable                                                                           |
| Hydrazine           | 302-01-2   | 0.5 tonne                                                                                 | 2 tonne                                                                                  |
| Water               | 7732-18-5  | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

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## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** Water endangering class = 3 (self classification)

| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                                                            |
|-----------|---------------------------------------|------------------------------------------------------------------------------------|
| Hydrazine | WGK3                                  | Krebserzeugende Stoffe - Class I : 0.05 mg/m <sup>3</sup><br>(Massenkonzentration) |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H226 - Flammable liquid and vapor  
H301 - Toxic if swallowed  
H311 - Toxic in contact with skin  
H330 - Fatal if inhaled  
H314 - Causes severe skin burns and eye damage  
H317 - May cause an allergic skin reaction  
H318 - Causes serious eye damage  
H350 - May cause cancer  
H400 - Very toxic to aquatic life  
H410 - Very toxic to aquatic life with long lasting effects

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

# SAFETY DATA SHEET

Hydrazine hydrate, 80% (Hydrazine, 51%)

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Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

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**Revision Summary** Next Revision Date - Oct 2028.

**This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.**

## Disclaimer

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**End of Safety Data Sheet**